

Renal Module Documentation

Folder-wise code map, screen-wise purpose, role-wise access, and presentation flow for the developed renal module.

Project: Plasmit Hospital HMS

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Main module route: `/renal`

Data mode: Static/mock workflow, production-style UI

Short explanation for presentation: Renal module is a role-aware clinical workspace for urine output, intake/output charting, drains/devices, renal labs, renal alerts, dialysis billing, reports, and final doctor sign-off. The module is designed so each user sees what they can do based on their role.

High-level workflow

1. Patient / Chart
Open existing renal chart or create a new renal chart.

2. Clinical Entry
Nurse/doctor records I/O, drains, urine output, and notes.

3. Review
Alerts, labs, orders, balance trends, and dialysis needs are reviewed.

4. Sign / Report
Doctor signs report, then report goes to EMR or billing handoff.

This document is generated for presentation and handover of the developed renal screens.

Folder-Wise Code Map

Folder / File	What is written here	Why it matters
<code>src/app/(app)/renal/page.tsx</code>	Route file for Renal Overview / Command Center.	Loads the main dashboard screen at <code>/renal</code> .
<code>src/app/(app)/renal/patients/page.tsx</code>	Route file for renal patient chart list.	Shows all active renal patient charts.
<code>src/app/(app)/renal/patients/[patientId]/page.tsx</code>	Dynamic route for individual patient renal chart.	Opens patient-specific renal workspace such as <code>/renal/patients/pat-002</code> .
<code>src/app/(app)/renal/fluid-balance/page.tsx</code>	Route file for all-patient fluid balance summary.	Used for quick 24-hour intake/output and cumulative balance review.
<code>src/app/(app)/renal/drains/page.tsx</code>	Route file for drains and devices.	Used for drain output, device status, and drain review workflow.
<code>src/app/(app)/renal/labs/page.tsx</code>	Route file for renal labs.	Used for creatinine, urea, electrolytes, eGFR, protein, flags, and trends.
<code>src/app/(app)/renal/reports/page.tsx</code>	Route file for reports and dialysis.	Used for report generation, sign-off queue, EMR handoff, and dialysis billing handoff.
<code>src/features/renal/renal-pages.tsx</code>	Main implementation file containing renal screens, tables, modals, filters, forms, workflows, and page components.	This is the main file where most renal UI logic is written.
<code>src/features/renal/renal-shared.tsx</code>	Shared renal UI and access helpers: role access, protected renal wrapper, status badges, balance badges, metric cards, clinical tables.	Keeps role access and repeated renal components consistent.
<code>src/data/renal.ts</code>	Static renal mock data and TypeScript models for charts, I/O rows, drains, labs, orders, alerts, and dialysis sessions.	Data source used by all renal screens during UI development.
<code>src/data/navigation.ts</code>	Sidebar menu entries for Renal module and submenus.	Controls sidebar labels and routes like Overview, Patient Charts, Fluid Balance, Drains, Labs, Reports.

Presentation point: Route files are lightweight. Actual screen development is centralized in `renal-pages.tsx`, shared access/UI is in `renal-shared.tsx`, and demo data is in `renal.ts`.

Sidebar and Screen List

Menu	Route	Screen purpose	Important features developed
Overview	/renal	Renal command center for active renal patients, alerts, and user actions.	Patients tab, Renal Alert Queue, My Actions, search/sort, new renal entry, alert acknowledgement, billing modal.
Patient Charts	/renal/patients	List of active inpatient renal charts.	Patient search, status/priority handling, open chart, centered review modal, create chart workflow.
Specific Patient Chart	/renal/patients/[patientId]	Complete patient-specific renal workspace.	Overview, I/O Chart, Labs & Orders, Notes & Sign-off tabs.
Fluid Balance	/renal/fluid-balance	All-patient 24-hour balance summary.	Total intake, output, balance, cumulative balance, dialysis review status.
Drains & Devices	/renal/drains	Drain/device output and condition monitoring.	Add drain reading modal, new/existing drain mode, device dropdown, site dropdown, review modal.
Renal Labs	/renal/labs	Renal investigation results and trend monitoring.	Search, filter, sort, update lab modal, review lab modal, critical/worsening metrics.
Reports	/renal/reports	Report generation, dialysis session review, EMR and billing handoff.	All/individual/selected patient report scope, balance report, sign-off queue, EMR handoff, dialysis billing.

Role-Wise Access

Access is controlled through `useRenalAccess()` in `src/features/renal/renal-shared.tsx`.

Role	Can view renal module?	I/O / drain entry	Doctor review / sign-off / orders	Lab update	Billing handoff
Super Admin	Yes	Yes	Yes	Yes	Yes
Hospital Admin	Yes	Yes	Yes	Yes	Yes
Doctor	Yes	Yes	Yes	Yes	No
Nurse	Yes	Yes	No	No	No
Lab Technician	Yes	No	No	Yes	No
Billing Executive	Yes	No	No	No	Yes
Management	Yes	Read only	Read only	Read only	Read only

My Actions tab

Action	Allowed role	When to use	After submit
Add I/O entry	Nurse / Doctor	During bedside hourly or shift charting.	Fluid balance table and renal chart get updated.
Acknowledge alert	Nurse / Doctor	When a renal rule needs acknowledgement or escalation.	Alert action is captured for audit and handoff.
Sign renal report	Doctor	After I/O, labs, alerts, and orders are reviewed.	Report becomes ready for signed clinical handoff.
Add renal order	Doctor	When follow-up action is required, such as repeat RFT or dialysis review.	Order appears in labs, orders, and sign-off checklist.
Update renal labs	Lab / Doctor	After a renal sample result is available.	Latest renal labs and abnormal flags get refreshed.
Prepare dialysis billing	Billing	When dialysis session status is billing pending.	Billing receives service rows without clinical chart edits.

Overview Screen Flow

Route: `/renal`

Patients tab

Search and sort renal charts. Open patient chart or quick review.

Renal Alert Queue

Severity grouped alerts.
Acknowledge alert from centered modal.

My Actions

Role-based action cards with clear locked/ready status.

Important modals from Overview

- **New renal entry:** supports existing chart or new chart, patient dropdown, entry type, bed/ward dropdown, fluid restriction dropdown, consultant/nephrologist dropdowns, and first entry draft.
- **Review:** opens centered renal chart review modal.
- **Acknowledge alert:** opens centered alert acknowledgement modal.
- **Prepare dialysis billing:** opens centered billing modal with patient scope dropdown, multi-session selection, package, payer status, priority, and estimate.

Start

Open `/renal` .

Select tab

Patients, Alerts, or My Actions.

Open action

Review, acknowledge, entry, billing, labs, or sign-off.

Submit

Action is staged in static workflow with toast feedback.

Individual Patient Chart

Route: /renal/patients/[patientId]

Tab / Section	Use	Main actions
Overview	Patient identity, bed, ward, renal status, intake/output summary, active drains, and risk context.	Print report, doctor sign-off, add I/O entry.
I/O Chart	Detailed intake, detailed output, and complete 24-hour table in one place.	Add intake entry, add output entry, view 24-hour balance table.
Labs & Orders	Renal lab trends, orders, and hourly balance graph.	Update labs, add renal order, review flagged results.
Notes & Sign-off	Final checklist before report sign-off.	Open doctor sign-off modal and sign renal report.

Presentation line: Individual patient chart is the main clinical screen. It keeps all renal information for one patient in a tab-wise structure so staff do not need to jump across multiple pages.

Drains, Labs, and Reports Screens

Drains & Devices

Feature	Purpose	Access
Add drain reading	Add output from existing drain or create new device using dropdowns for drain/device and site.	Nurse / Doctor
Review drain	Review status, 24h output, character, concern, related alerts, and next action.	Nurse / Doctor can record; other allowed roles view.

Renal Labs

Feature	Purpose	Access
Search / filter / sort	Find patient labs by flag, trend, potassium, creatinine, or patient.	All renal allowed roles view.
Update labs	Enter creatinine, urea, sodium, potassium, eGFR, urine protein, flag, and trend.	Lab Technician / Doctor / Admin
Review lab	Review critical or worsening results with related alerts and orders.	Doctor / Lab / Admin

Reports

Report feature	Purpose	Patient scope
24h fluid balance report	Generate printable intake/output/cumulative balance summary.	All active, individual patient, or selected patients.
Doctor sign-off queue	Check open alerts, pending orders, and latest labs before signing.	All active, individual patient, or selected patients.
EMR handoff	Send FHIR-aligned renal report bundle to EMR workflow.	All active, individual patient, or selected patients.
Dialysis billing handoff	Select dialysis sessions, choose package/payer/priority, estimate amount, and handoff to billing.	All patients, individual patient, or selected patients via dropdown/scope.

FHIR / HL7-Friendly Mapping

The UI is structured so renal data can map to healthcare interoperability resources later.

FHIR resource	Renal module data	Screen examples
Patient	Patient identity, UHID, demographics.	Overview, Patient Charts, Reports.
Encounter	Admission, bed, ward, active inpatient context.	Patient chart header, report scope.
Observation	Intake, output, hourly urine, fluid balance, cumulative balance.	I/O Chart, Fluid Balance.
Device	Foley catheter, drains, NG tube, chest drain, device status.	Drains & Devices, patient detail.
DiagnosticReport	Renal labs: creatinine, urea, sodium, potassium, eGFR, protein.	Renal Labs, Labs & Orders.
ServiceRequest	Doctor renal orders: repeat RFT, dialysis review, potassium watch.	Labs & Orders, Add renal order.
AuditEvent / Provenance	Who viewed, entered, acknowledged, signed, or handed off records.	My Actions, Reports, EMR handoff.

Important: Current implementation uses mock/static data for UI development. Backend persistence, real PDF generation, digital signature, and real EMR/FHIR API integration can be added in the next production phase.

Presentation Script

1. **Start from sidebar:** Show Renal menu and submenus: Overview, Patient Charts, Fluid Balance, Drains & Devices, Renal Labs, Reports.
2. **Explain role banner:** Screen changes based on selected role. Nurse sees charting actions, Doctor sees review/sign-off, Lab sees lab update, Billing sees billing handoff.
3. **Overview:** Show Patients, Renal Alert Queue, and My Actions. Explain that My Actions tells each user what they can do now.
4. **New renal entry:** Show dropdown-based patient/chart selection. Explain existing chart vs new chart scenario.
5. **Patient detail:** Open one patient and show tabs: Overview, I/O Chart, Labs & Orders, Notes & Sign-off.
6. **Drains:** Show add drain reading modal and review modal. Explain device/site dropdowns.
7. **Labs:** Show search/filter/sort, update lab modal, review lab modal.
8. **Reports:** Show report scope: all patients, individual patient, selected patients. Explain EMR handoff and dialysis billing.
9. **End with access:** Explain role matrix and why locked buttons protect clinical workflows.

One-minute summary

Renal module is a complete workflow from bedside I/O entry to doctor sign-off and reporting. Nurse enters I/O and drains, Lab updates renal investigations, Doctor reviews alerts/orders and signs report, Billing handles dialysis billing, and Management can review read-only dashboards. Every action opens in a focused centered modal and is designed for quick use in hospital operations.